Business Requirements

IT Ticket Auto-Classification System - Table Explanation

1. Description

Point	Explanation
Problem	IT tickets are tagged manually with Category and Subcategory. This is slow, inconsistent and error-prone.
Solution	A system that reads the caller's issue description and automatically classifies the ticket, emails the caller, and stores the data in a standard format.
Goal	Faster routing and resolution, consistent data, less manual effort, and a solution that is easy to maintain and scale.

2. Business Requirements - Detailed Explanation

The system must:

No.	Requirement	What it means	Example	Benefit
1	Automatically classify IT tickets based on the issue description	The system reads the free-text description and decides the type of issue using keyword rules, an ML model or AI.	"Cannot connect to office Wi-Fi" is identified as a network issue.	No agent has to read and tag each ticket; consistent results.
2	Assign both Category and Subcategory without manual intervention	Two levels are filled automatically at ticket creation: Category (broad area) and Subcategory (specific issue).	"Outlook keeps crashing" becomes Category: Software, Subcategory: Email Client.	Tickets reach the right team faster.
3	Support dependent choice logic between Category and Subcategory	Subcategory options depend on the chosen Category. Only valid combinations are allowed.	Hardware can have Printer, but never VPN.	Clean, accurate data; no wrong combinations.
4	Send an automated email notification to the caller upon ticket creation	As soon as the ticket is created, the caller gets a confirmation email automatically.	Email with ticket number, summary, category and status.	Caller is reassured; fewer follow-up queries.

No.	Requirement	What it means	Example	Benefit
5	Store ticket information in a structured and standardized format	Every ticket is saved with the same fields and naming conventions.	Ticket ID, Caller, Description, Category, Subcategory, Status, Created Date.	Easy search, filtering, reporting and analytics.
6	Ensure easy maintenance and future scalability	Lists and rules are kept in configuration, design is modular, and it handles growth.	Add a new Category by editing a table, not code.	Low maintenance cost; ready for higher volume.

3. Dependent Choice Logic - Category vs Subcategory

Category (broad area)	Valid Subcategories (specific issue)	Sample ticket description
Network	Wi-Fi, VPN, LAN	VPN is not connecting from home
Hardware	Laptop, Printer, Monitor	Printer is not printing
Software	Email Client, Operating System, Installation	Outlook keeps crashing
Access / Security	Password Reset, Account Locked, Permissions	My account is locked

4. Ticket Data Structure (Standardized Storage)

Field	Type	Description	Example
Ticket ID	Text (unique)	Unique number generated for each ticket	INC0001023
Caller Name / Email	Text	Person who raised the ticket; used for notification	ravi@company.com
Description	Long text	Issue described by the caller	Cannot connect to office Wi-Fi
Category	Choice	Auto-assigned broad area	Network
Subcategory	Choice (depends on Category)	Auto-assigned specific issue	Wi-Fi
Status	Choice	Current state of the ticket	New
Created Date	Date and time	When the ticket was created	30-09-2026 10:15

5. Email Notification Content

Email item	Purpose
Ticket number	Lets the caller track or refer to the ticket

Email item	Purpose
Issue summary	Confirms the problem was understood correctly
Category and Subcategory	Shows how the ticket was classified
Status and next steps	Tells the caller what to expect next